Lab 1 — Getting Started with QGIS

Introductory lab · Software setup and interface overview · QGIS 3.44 LTR

Goals of this lab

- Install QGIS 3.44 LTR correctly on your own machine.
- Recognise the main parts of the QGIS interface.
- Be ready to load your first layers in the next lab.

1. Why QGIS?

Free of charge	Installing and using QGIS costs nothing — no licence fee, no subscription.
Open source	You can inspect it, extend it, or sponsor a new feature instead of just waiting for one.
Backed by OSGeo	An official project of the Open Source Geospatial Foundation, aligned with open geospatial standards.
Actively developed	New tools and improvements arrive continuously.
Well documented	Extensive manuals, a training manual, and a large user community.
Cross-platform	Runs on Windows, macOS and Linux.

2. Layers

Everything you open in QGIS is a **layer**: one dataset of one kind of object — roads, schools, a satellite image. Layers stack on top of each other like transparent sheets, and they are listed in the Layers panel on the left:

- Unchecking the box next to a layer hides it from the map.
- Their order in the list is their order on the map — upper layers cover lower ones.

3. Installing QGIS

Official website: `qgis.org` → **Download**.

1. **Choose the LTR version — QGIS 3.44 LTR.** The page offers the newest release first; look for the button labelled *LTR* or *Long Term Release*. Do not download version 4.

2. **Pick the installer for your system.**
   Windows: the *Standalone Installer* (about 1.5 GB).
   macOS: the `.dmg` file.

3. **Run the installer.** Click *Next* through every screen and keep the default install location.
   If Windows shows a SmartScreen warning, click *More info* → *Run anyway*.
   On macOS, drag the QGIS icon into the *Applications* folder.

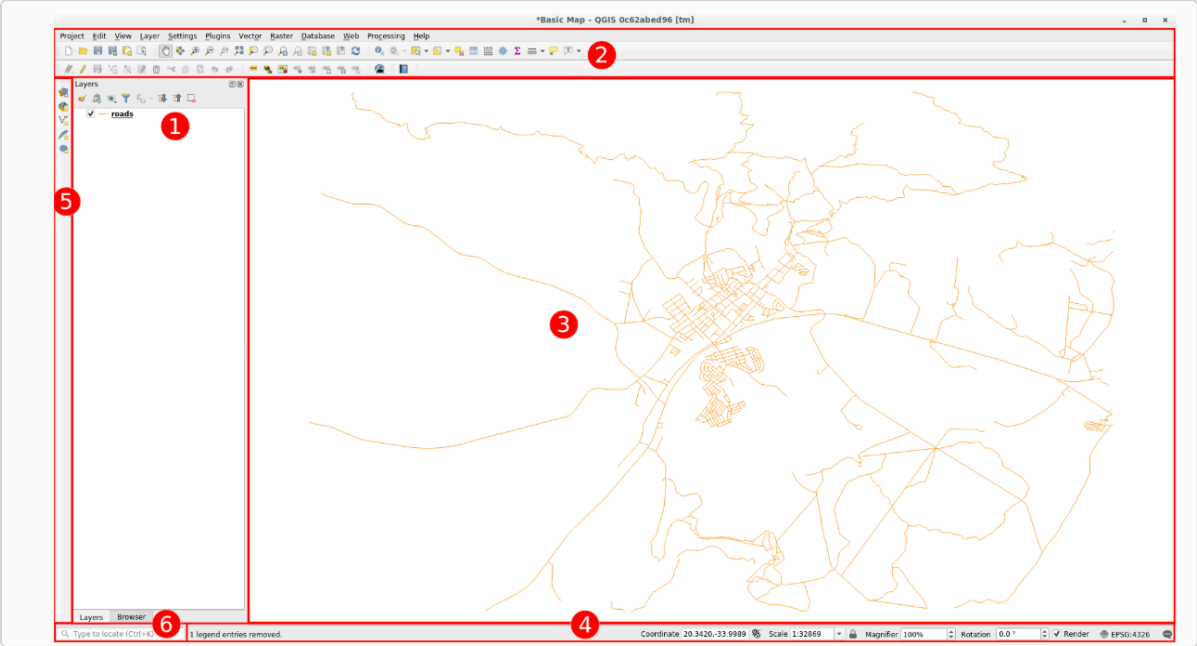
4. **Open the program.** The Start menu will show several icons — open the one called **QGIS Desktop 3.44**. Ignore the others for now.
   On macOS, the first time only: right-click the icon → *Open* → *Open* again.

5. **Check that it works.** QGIS opens with an empty map, the Layers panel on the left, and `EPSG:4326` in the bottom-right corner.

Important

- Everyone must install the **same version (3.44 LTR)**, so that menus and buttons are in the same place for all of us.
- Why LTR and not the newest release? An LTR receives bug fixes only, with no surprise interface changes mid-term, and almost every tutorial and textbook you will find online still targets the 3.x series.
- Make sure you have at least **4 GB of free disk space**.
- Do not install QGIS into a folder path containing non-English characters.
- Keep the interface language in **English** (`Settings → Options → General`) so it matches the documentation.

4. An Overview of the Interface



The QGIS user interface — source: QGIS Training Manual 3.44

#	Element	What it does

1	Layers List / Browser Panel	Lists every layer currently available. Expand a layer to see how it is drawn; hover over it for its name, geometry type, coordinate system and file path; right-click it for a long menu of options. The *Browser* tab beside it lets you navigate files, databases and web services, and mark folders as Favorites.
2	Toolbars	Grouped buttons for frequent tasks. Toolbars can be shown or hidden from `View → Toolbars`; every tool also remains reachable through the menus.
3	Map Canvas	Where the map is drawn and where you interact with the layers — zoom, pan, select features.
4	Status Bar	Information about the current map: cursor coordinates, map scale, rotation, and the project's coordinate reference system (`EPSG`).
5	Side Toolbar	Buttons for loading layers and creating new ones. Like any toolbar, it can be moved.
6	Locator Bar	Search box reaching almost everything in QGIS — layers, features, algorithms, bookmarks. Shortcut: `Ctrl + K` .

Before the next lab

- Install QGIS at home and confirm that it opens — the download is large and takes time on lab Wi-Fi.
- If the installation fails, take a photo of the error message and bring it to the start of the next lab.
- Next lesson: loading your first layers into a map.
